

EXPERIMENT – 01

AIM: To study various communication media used in computer networks and understand the structure and working of straight-through and cross-wired Ethernet cables.

OBJECTIVE:

- To understand different types of transmission media
- To learn how data travels through wired communication channels
- To understand the internal wiring standards of Ethernet cables
- To differentiate between straight-through and cross-over cables

THEORY:

Communication Media

Communication media refers to the physical or wireless path through which data travels from one device to another in a network. It plays a crucial role in determining the speed, reliability, and efficiency of communication.

Communication media is broadly classified into:

A. Guided Media (Wired Communication)

These are physical paths through which signals are transmitted.

1. Twisted Pair Cable

- Most commonly used networking cable
- Consists of pairs of insulated copper wires twisted together
- Twisting reduces electromagnetic interference
- Two types:
 - **UTP (Unshielded Twisted Pair)** – widely used in LANs
 - **STP (Shielded Twisted Pair)** – has shielding for extra protection

Used in Ethernet networks (LANs).

2. Coaxial Cable

- Contains a central conductor, insulating layer, metallic shield, and outer cover
- Provides better protection from interference than twisted pair
- Used in cable TV and older network systems

3. Optical Fiber Cable

- Uses light signals instead of electrical signals
- Very high speed and bandwidth
- Immune to electromagnetic interference
- Used in long-distance communication and high-speed networks

B. Unguided Media (Wireless Communication)

These do not require a physical connection.

1. Radio Waves

- Used in Wi-Fi and broadcasting
- Can travel long distances

2. Microwaves

- Used in satellite communication
- Requires line-of-sight transmission

3. Infrared

- Used in short-range communication (e.g., remote controls)

Ethernet Cable Wiring Concepts

Ethernet cables (usually UTP) use **RJ-45 connectors** and contain **8 wires (4 pairs)**. These wires follow specific color codes defined by standards.

Two common wiring standards:

- **T568A**
- **T568B**

Straight-Through Cable

Definition

A straight-through cable is a type of Ethernet cable in which the wiring pattern is the same on both ends (either T568A–T568A or T568B–T568B).

Working Principle

- Each pin connects directly to the same pin number on the other end
- Transmit and receive signals remain aligned as required by different device types

Usage

Used to connect **different types of devices**, such as:

- Computer to Switch
- Computer to Router
- Switch to Router

Cross-Over Cable (Cross-Wired Cable)

Definition

A cross-over cable is an Ethernet cable where the transmit and receive pairs are swapped between the two ends (T568A on one end and T568B on the other).

Working Principle

- Transmit (TX) pins on one end connect to receive (RX) pins on the other end
- Enables direct communication between similar devices

Usage

Used to connect **same types of devices**, such as:

- Computer to Computer
- Switch to Switch (older devices)
- Router to Router

Implementation of Crimping Tool (Conceptual)

- The crimping tool is used during the **termination of Ethernet cables** with RJ-45 connectors
- First, the cable is **prepared by removing outer insulation** to expose internal wires
- The wires are then **arranged in a specific color order** (T568A or T568B standard)
- The arranged wires are inserted into the **RJ-45 connector** in proper alignment
- The connector is placed into the **crimping tool slot**
- By pressing the tool handles, **metal pins inside the connector are forced into the wires**
- This creates a **secure electrical and physical connection**
- The tool also tightens the connector around the cable to ensure **firm holding and durability**
- Proper crimping results in **stable network communication without data loss**

EXPERIMENT – 02

AIM: To study different types of networks devices – Routers, Switches, Hub, Bridge, Repeater, gateway how they work and where are they used.

THEORY:

In a computer network, different devices are used to connect systems, manage data flow, and ensure smooth communication between multiple devices. Each device operates at a specific level and performs a distinct function, depending on the requirement of the network.

- **HUB:** A hub is one of the simplest networking devices. It works by receiving data from one device and broadcasting it to all other connected devices without any filtering or decision-making. Since it does not check the destination address, all devices receive the data, leading to unnecessary traffic and reduced efficiency. Hubs are mainly used in very basic or older network setups.
- **SWITCH:** A **switch** is an advanced version of a hub and is widely used in modern local area networks. Unlike a hub, a switch intelligently forwards data only to the intended device by using MAC addresses. It maintains a table of connected devices and sends data to the correct port, which improves network efficiency and reduces collisions.
- **ROUTER:** A router is used to connect different networks together, such as connecting a local network to the internet. It works by analyzing IP addresses and determining the best path for data to travel from source to destination. Routers also provide features like network address translation (NAT), security, and traffic management, making them essential in both home and enterprise networks.
- **BRIDGE:** A bridge is used to divide a large network into smaller segments. It works by filtering traffic based on MAC addresses and forwarding data only when necessary. Bridges help reduce network congestion by limiting unnecessary data transmission between segments.

Although less commonly used today, their function is largely integrated into switches.

- **REPEATER:** A repeater is a device used to extend the range of a network. As signals travel over long distances, they become weak. A repeater receives the weakened signal, amplifies or regenerates it, and retransmits it to its destination. It does not filter or modify data, but simply strengthens the signal.
- **GATEWAY:** A gateway acts as a connecting point between different types of networks that may use different protocols. It performs protocol conversion so that communication can occur between incompatible systems. Gateways are commonly used to connect networks with different architectures, such as connecting a local network to external networks or services.

Overall, these devices work together to ensure efficient data transmission, proper routing, signal strength maintenance, and communication between different parts of a network. Their selection and use depend on the size, complexity, and requirements of the network.

EXPERIMENT – 03

AIM: To study basics networking commands.

THEORY:

Basic networking commands are used to **check connectivity, diagnose network issues, and gather network information**. These commands are executed through the command line interface.

Ping

- Used to check **connectivity between two devices**
- Sends **ICMP echo request** and waits for reply
- Measures **response time (latency)**
- Helps identify whether a host is **reachable or not**

ipconfig

- Displays **IP configuration** of a system
- Shows details like **IP address, subnet mask, default gateway**
- Used to verify network settings
- Helps in troubleshooting IP-related issues

tracert (Traceroute)

- Shows the **path taken by data packets** to reach destination
- Displays all **intermediate routers (hops)**
- Helps identify **where delay or failure occurs**

netstat

- Displays **active network connections**
- Shows **open ports and protocols**
- Useful for monitoring network activity
- Helps detect **unauthorized connections**

nslookup

- Used to find **IP address of a domain name**
- Queries **DNS server**
- Helps in troubleshooting **DNS-related issues**

hostname

- Displays the **name of the current system**
- Helps in identifying devices in a network

arp

- Displays and manages **ARP table (IP to MAC mapping)**
- Helps understand how devices are mapped in a LAN

getmac

- Displays the **MAC address** of the system
- Useful for identifying devices in a network
- Helps in **network configuration and troubleshooting**

pathping

- Combines features of **ping and tracert**
- Shows **path to destination** and packet loss at each hop
- Useful for **detailed network analysis**

route

- Displays and manages the **routing table**
- Shows how data is routed in the network
- Helps in understanding and troubleshooting **routing issues**

EXPERIMENT – 04

AIM: Configure and implement a network using Packet Tracer. Creating a First Network.

Introduction

Cisco Packet Tracer is a network simulation software developed by Cisco Systems that allows users to design, configure, and test computer networks in a virtual environment.

Creating a first network in Packet Tracer helps beginners understand the basic structure of a computer network, including how devices such as computers, switches, and routers communicate with each other. It also helps in understanding the process of assigning IP addresses and testing connectivity between devices.

Concept of a Computer Network

A computer network is a collection of interconnected devices such as computers, switches, routers, and servers that communicate with each other to share data, resources, and services. Networks enable communication between different devices using networking protocols such as TCP/IP.

Cisco Packet Tracer

Cisco Packet Tracer provides a graphical interface where users can drag and drop networking devices such as PCs, switches, routers, hubs, and cables into a workspace to create network topologies. It also provides configuration tools to assign IP addresses, configure devices, and simulate network communication.

It helps users understand networking concepts such as:

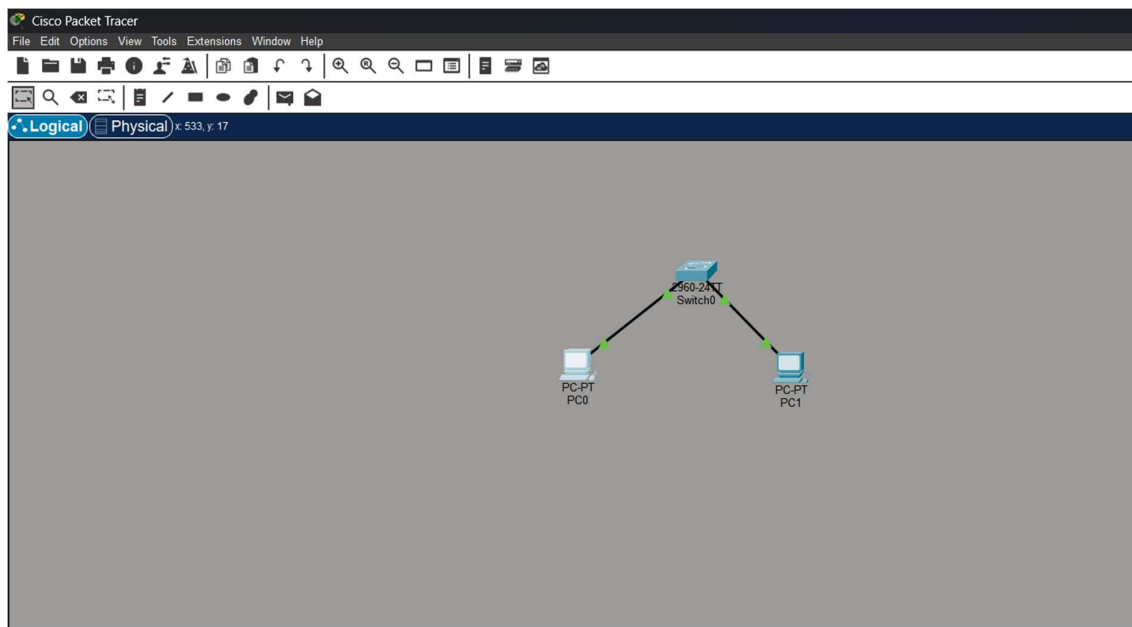
- Network topology
- Device configuration

- IP addressing
- Network connectivity
- Troubleshooting

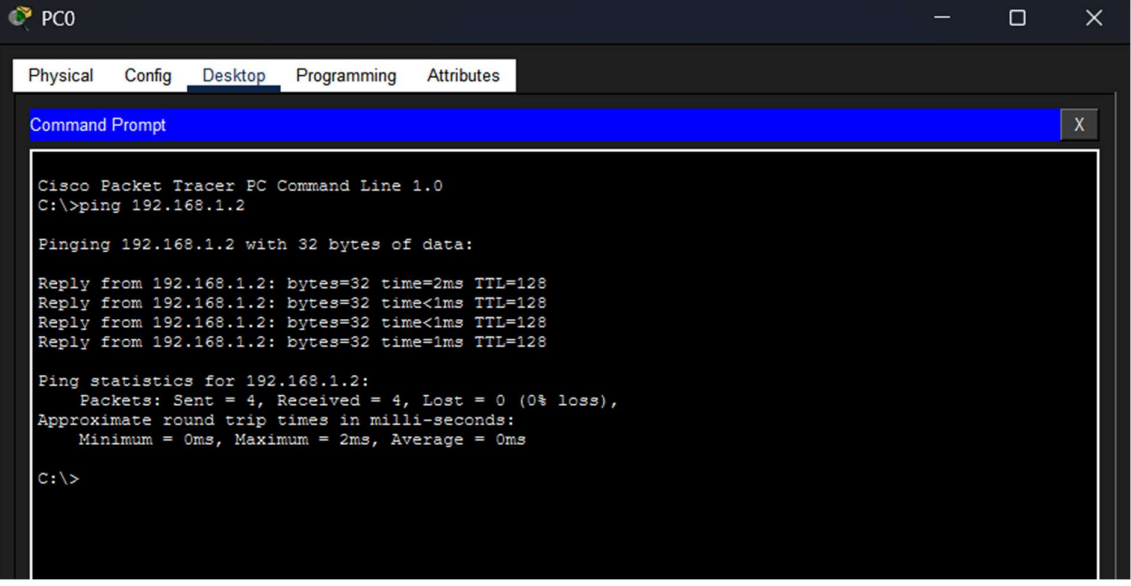
Network Topology

Network topology refers to the physical or logical arrangement of devices in a network. In the first network created using Packet Tracer, a simple topology is usually used where two PCs are connected through a switch.

This type of topology allows both computers to communicate with each other within the same network.



OUTPUT:



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the command prompt is as follows:

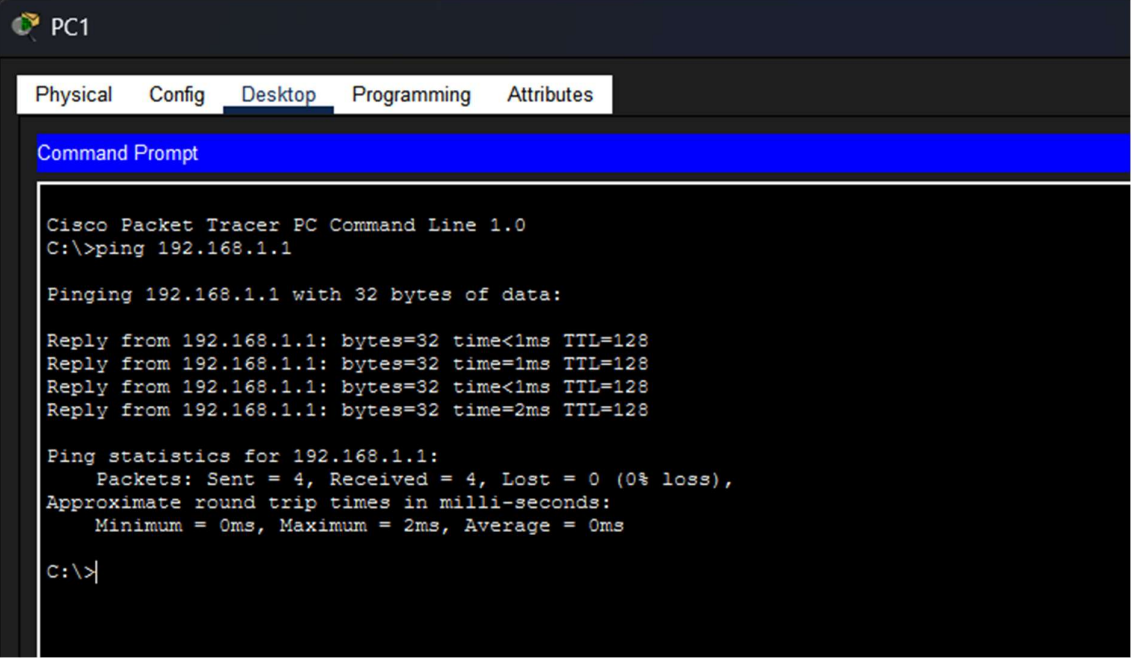
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```



The screenshot shows the 'PC1' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the command prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>|
```

IP Addressing

For devices to communicate in a network, each device must have a unique **IP address**. An IP address is a numerical label assigned to each device connected to a network.

An IP address consists of two parts:

1. **Network ID** – identifies the network.
2. **Host ID** – identifies the device within that network.

Both devices must belong to the same network and have the same **subnet mask**, usually **255.255.255.0**.

Network Communication

After configuring the devices and assigning IP addresses, connectivity between devices can be tested using networking commands such as **ping**.

The **ping command** sends packets from one device to another to check whether the destination device is reachable. If the connection is successful, the destination device sends a reply, confirming that communication between the devices is working properly.

Importance of Creating a First Network

Creating a first network in Packet Tracer helps students:

- Understand basic networking concepts.
- Learn how to connect networking devices.
- Configure IP addresses on devices.
- Test network connectivity.
- Gain practical knowledge of network simulation.

It also provides a foundation for learning advanced networking topics such as **routing, switching, and network security**.

EXPERIMENT – 05

AIM: Configure network devices, such as hubs and switches within a network topology using Packet Tracer software.

Introduction

A computer network is a system in which multiple devices such as computers, printers, and servers are connected together to communicate and share resources. Networking devices like **hubs and switches** are used to connect multiple devices within a local network. Cisco Packet Tracer is a **network simulation tool developed by Cisco Systems** that allows users to design, configure, and simulate network topologies in a virtual environment. It helps students understand how networking devices work and how networks are implemented.

Hub

A **hub** is a basic networking device used to connect multiple computers in a network. It works at the **Physical Layer (Layer 1) of the OSI model**. A hub receives data from one device and broadcasts it to all other connected devices in the network, regardless of the destination.

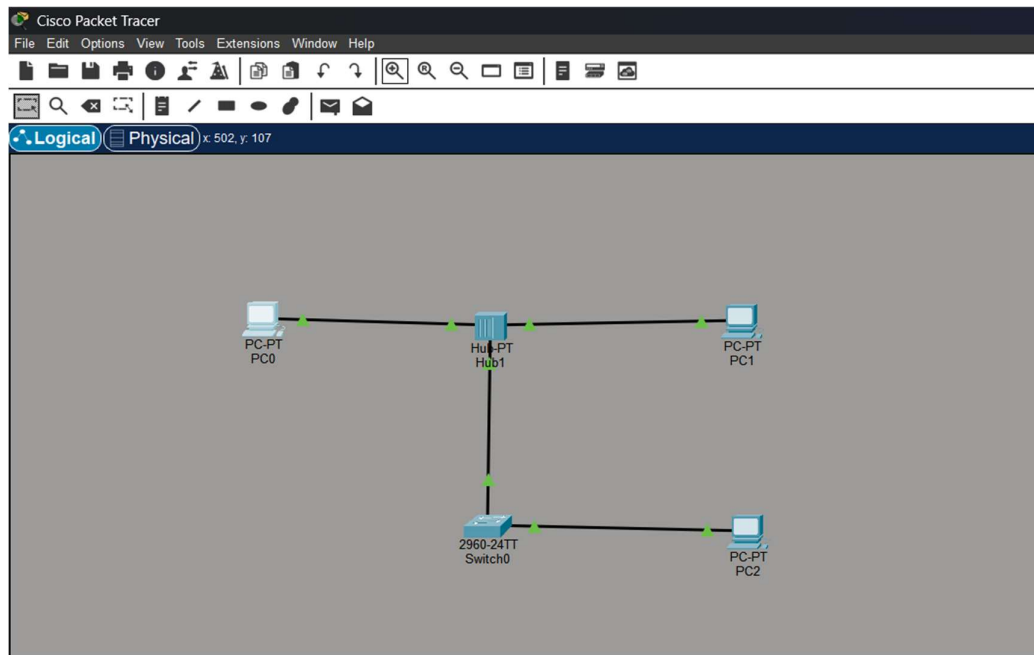
Features of Hub:

- Works on the Physical Layer of the OSI model
- Broadcasts data to all connected devices
- Does not filter or manage network traffic
- Simple and inexpensive device
- Used in small or basic networks

Switch

A **switch** is a more advanced networking device used to connect devices in a Local Area Network (LAN). It works at the **Data Link Layer (Layer 2) of the OSI model**. Unlike a hub, a switch sends data only to the specific device for which the data is intended.

Features of Switch:

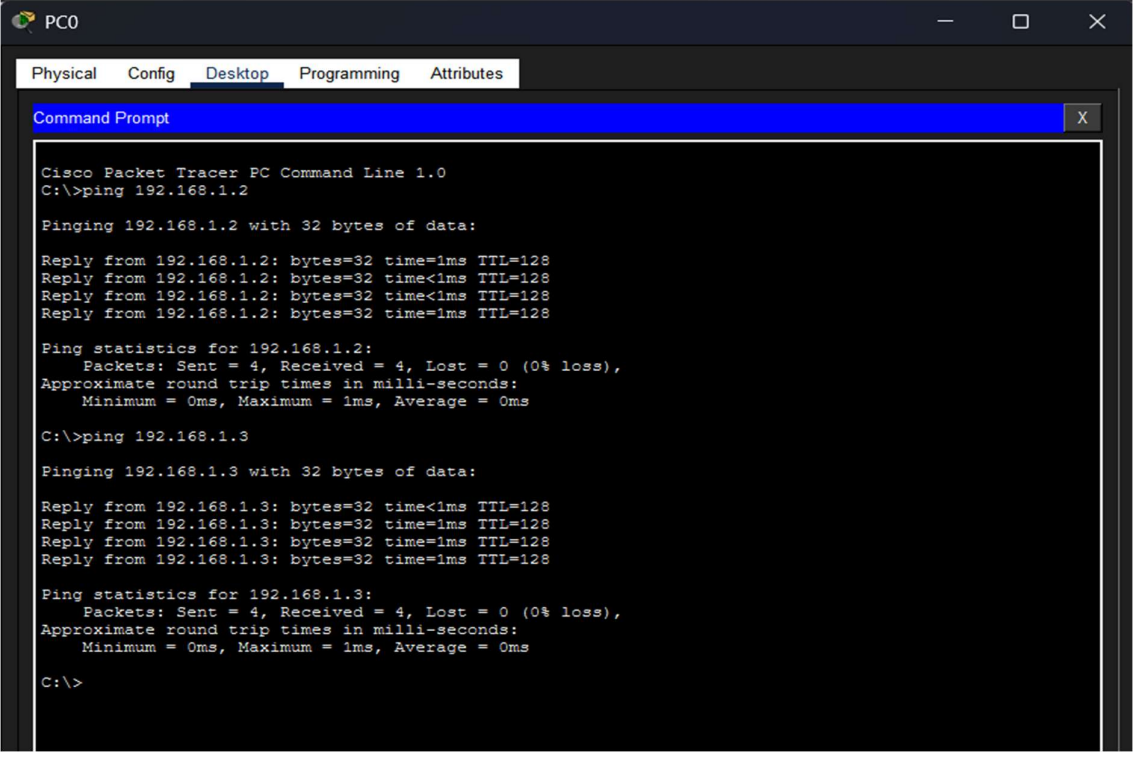


- Operates at the Data Link Layer of the OSI model
- Maintains a MAC address table
- Sends data only to the intended device
- Reduces network traffic and collisions
- Provides better performance than hubs

Network Topology

Network topology refers to the **arrangement or layout of devices in a network**. In a simple topology using a hub or switch, multiple computers are connected to a central networking device.

OUTPUT:



The screenshot shows the Command Prompt window for PC0 in Cisco Packet Tracer. The window has a title bar with 'PC0' and standard minimize, maximize, and close buttons. Below the title bar are tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The Command Prompt itself has a blue title bar with 'Command Prompt' and a close button. The text inside the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

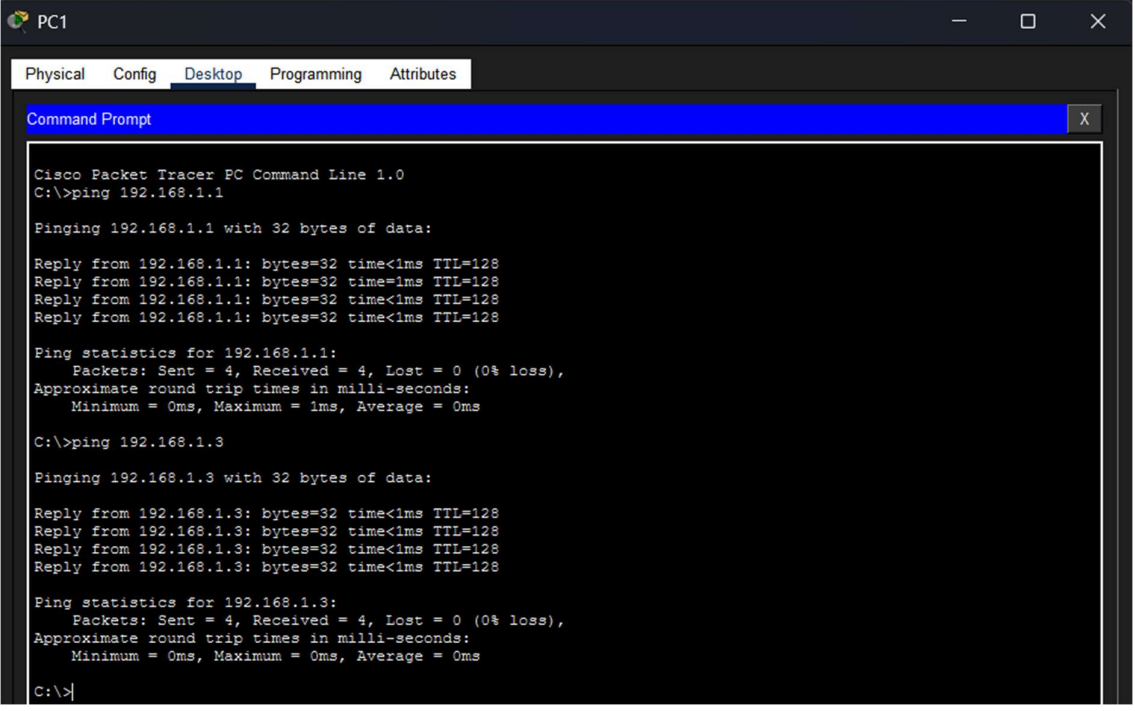
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```



The screenshot shows the Command Prompt window for PC1 in Cisco Packet Tracer. The window has a title bar with 'PC1' and standard minimize, maximize, and close buttons. Below the title bar are tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The Command Prompt itself has a blue title bar with 'Command Prompt' and a close button. The text inside the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

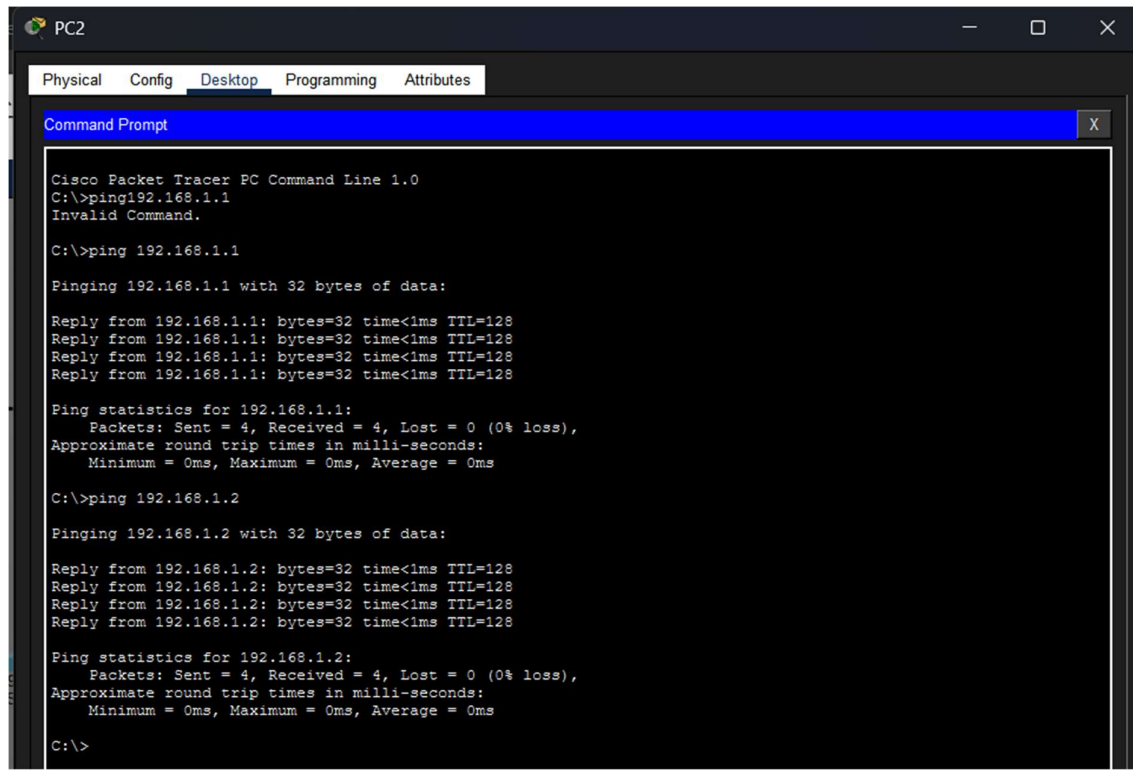
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping192.168.1.1
Invalid Command.

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

In this topology, the hub or switch acts as a central device that connects all computers and allows them to communicate with each other.

Working Principle

When devices are connected through a hub or switch, each device is assigned a unique IP address within the same network. When one device sends data to another device, the networking device (hub or switch) forwards the data across the network.

In the case of a hub, the data is broadcast to all connected devices. In contrast, a switch identifies the destination device using the MAC address and sends the data only to the correct device.

Importance

Configuring hubs and switches in Packet Tracer helps students understand the basic functioning of networking devices and the structure of local networks. It also helps in learning how devices communicate and how network connectivity is established.

EXPERIMENT – 06

AIM: Configure network topology and implement static routing using Packet Tracer Software.

Introduction

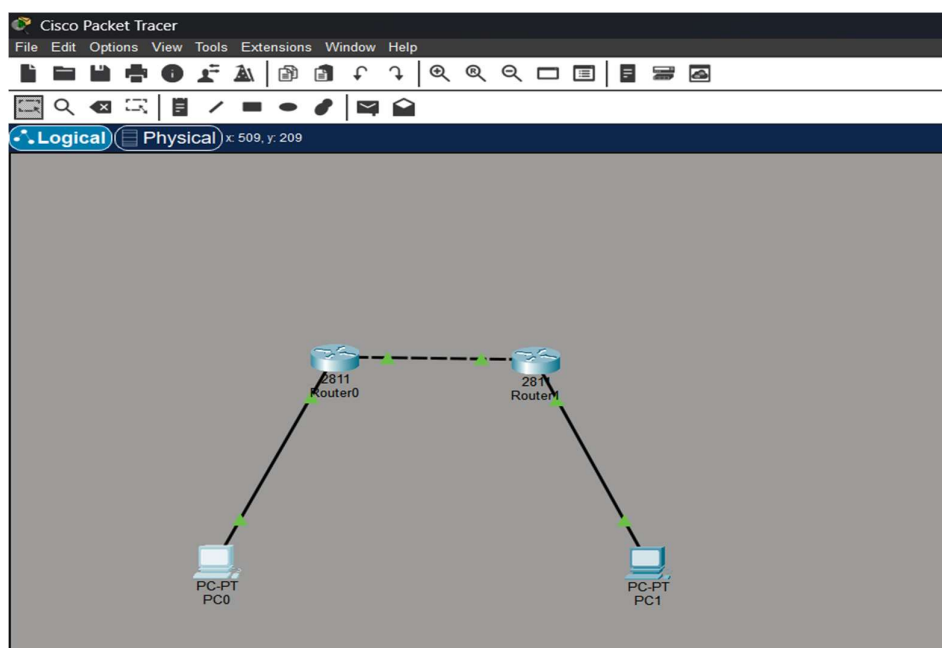
A computer network is a group of interconnected devices such as computers, switches, and routers that communicate with each other to share information and resources. To design and test networks practically, networking professionals and students use simulation tools like Cisco Packet Tracer.

In networking, when multiple networks are connected through routers, a routing mechanism is required to transfer data from one network to another. One of the simplest routing methods used in small networks is static routing.

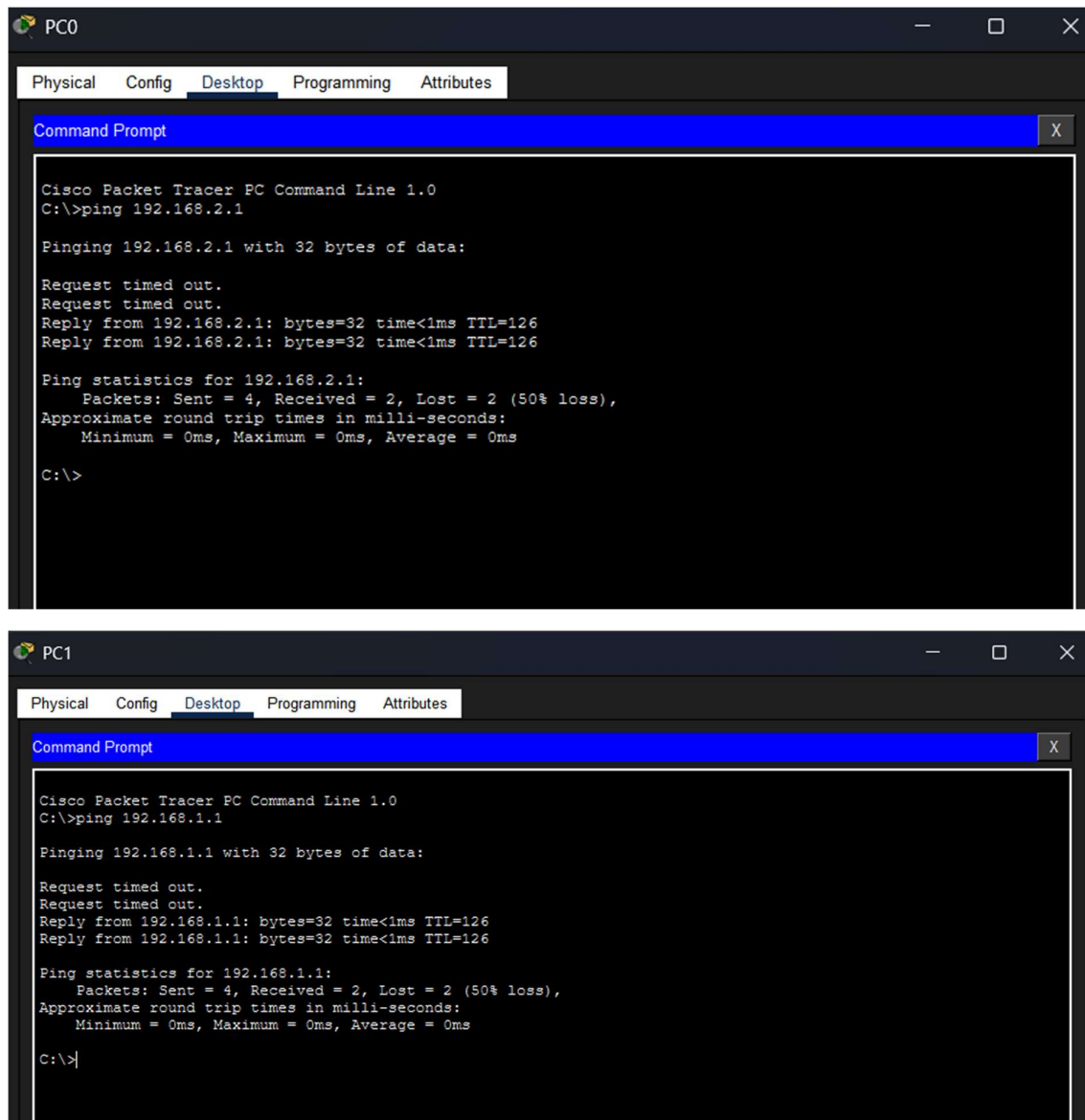
Network Topology

Network topology refers to the arrangement or layout of networking devices and connections in a network. It defines how devices such as computers, switches, and routers are connected to each other.

In Packet Tracer, users can design network topologies by placing devices in the workspace and connecting them using communication cables. A basic topology for static routing generally includes two or more routers connected to different networks.



OUTPUT:



The image displays two screenshots of the Cisco Packet Tracer PC Command Line interface. The top screenshot is for PC0, showing a ping command to 192.168.2.1. The output indicates that the ping failed with a 50% loss (2 out of 4 packets received). The bottom screenshot is for PC1, showing a ping command to 192.168.1.1. The output also indicates a 50% loss (2 out of 4 packets received).

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.1: bytes=32 time<1ms TTL=126
Reply from 192.168.1.1: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

In this topology:

- PCs act as end devices.
- Switches connect computers within the same local network.
- Routers connect different networks and enable communication between them.

Static Routing

Static routing is a routing technique in which routes are manually configured in the router by the network administrator. In static routing, the administrator

specifies the path that data packets must follow to reach a particular destination network.

Unlike dynamic routing, static routing does not automatically update routing information. Instead, routes must be manually entered and maintained by the administrator.

Working of Static Routing

When a data packet is sent from one network to another, the router checks its **routing table** to determine the path to the destination network. In static routing, the routing table contains manually configured routes that guide the router in forwarding packets.

Limitations of Static Routing

- Not suitable for large networks
- Requires manual configuration for every route
- Difficult to maintain if the network topology changes
- No automatic route updates

Importance of Static Routing in Networking

Static routing is commonly used in small networks or in situations where the network topology is stable and rarely changes. It helps administrators control the exact path that data packets follow within the network.

Learning static routing also provides a strong foundation for understanding advanced routing concepts such as **dynamic routing protocols** (RIP, OSPF, and EIGRP).

EXPERIMENT – 07

AIM: Configure network topology and implement dynamic routing protocol such as RIP, OSPF etc. using Packet Tracer.

Introduction

A computer network is a collection of interconnected devices such as computers, switches, and routers that communicate with each other to exchange data and share resources. To design, configure, and test networks practically, networking students and professionals use simulation software like Cisco Packet Tracer.

When multiple networks are connected together, routers are used to transfer data packets between different networks. For efficient communication between networks, routing protocols are required. One commonly used method is dynamic routing, where routers automatically exchange routing information and determine the best path for data transmission.

Dynamic routing protocols such as RIP (Routing Information Protocol) and OSPF (Open Shortest Path First) help routers learn network paths automatically without manual route configuration.

Network Topology

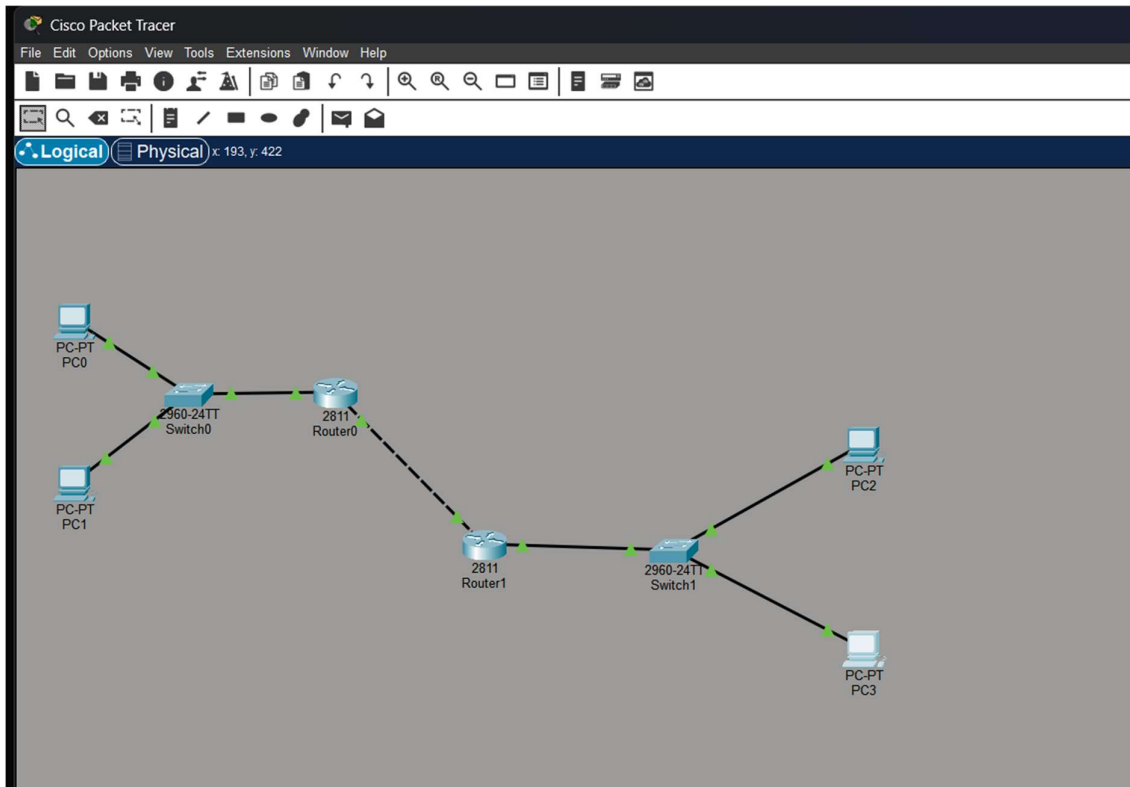
Network topology refers to the physical or logical arrangement of networking devices and communication links within a network. It defines how devices such as computers, switches, and routers are interconnected.

In Cisco Packet Tracer, users can create network topologies by placing devices into the workspace and connecting them using appropriate cables. A basic topology for dynamic routing generally consists of two or more routers connected to separate local area networks (LANs).

In this topology:

- PCs act as end devices used for communication.
- Switches connect multiple computers within the same LAN.
- Routers connect different networks together.

- Dynamic routing protocols allow routers to exchange routing information automatically.



```

PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=7ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.20

Pinging 192.168.2.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.20: bytes=32 time<1ms TTL=126
Reply from 192.168.2.20: bytes=32 time<1ms TTL=126
Reply from 192.168.2.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

```

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.20

Pinging 192.168.2.20 with 32 bytes of data:

Reply from 192.168.2.20: bytes=32 time<1ms TTL=126
Reply from 192.168.2.20: bytes=32 time=12ms TTL=126
Reply from 192.168.2.20: bytes=32 time<1ms TTL=126
Reply from 192.168.2.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>|
```

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

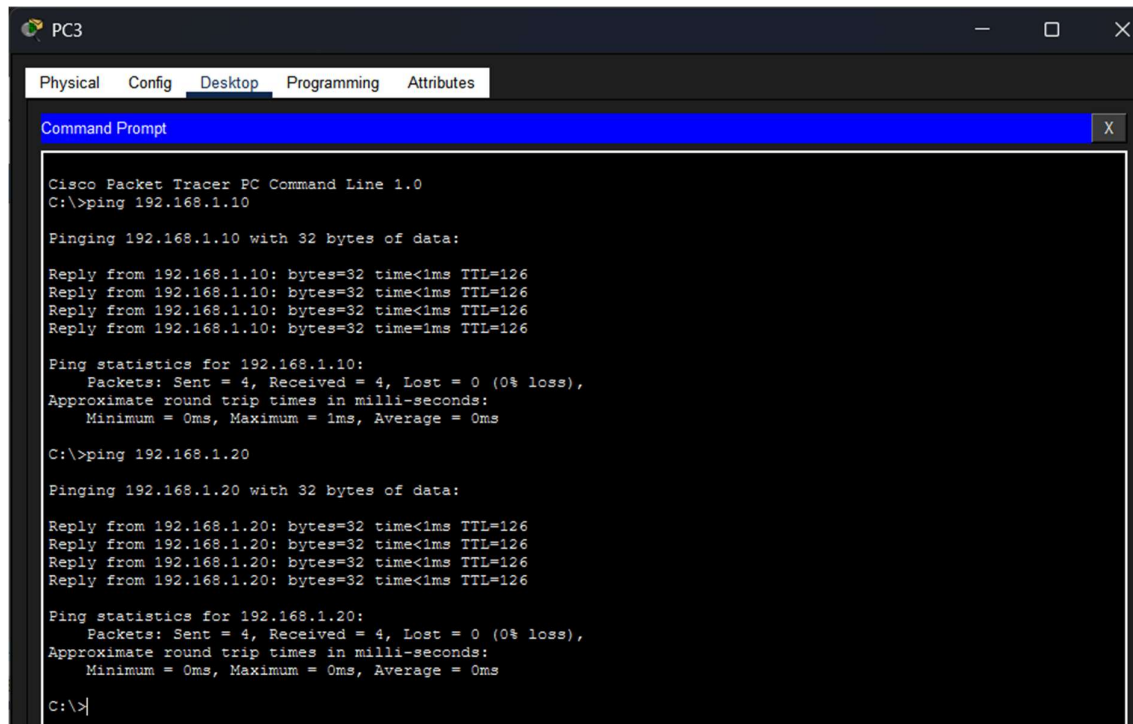
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=126
Reply from 192.168.1.20: bytes=32 time=1ms TTL=126
Reply from 192.168.1.20: bytes=32 time=1ms TTL=126
Reply from 192.168.1.20: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```



```
PC3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=126
Reply from 192.168.1.20: bytes=32 time<1ms TTL=126
Reply from 192.168.1.20: bytes=32 time<1ms TTL=126
Reply from 192.168.1.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Dynamic Routing

Dynamic routing is a routing method in which routers automatically exchange routing information with one another using routing protocols. Unlike static routing, routes do not need to be manually configured for every destination network.

Dynamic routing protocols help routers:

- Discover remote networks
- Update routing tables automatically
- Select the best path for data transmission
- Adapt to changes in network topology

Some commonly used dynamic routing protocols are:

- RIP (Routing Information Protocol)
- OSPF (Open Shortest Path First)
- EIGRP (Enhanced Interior Gateway Routing Protocol)

In this practical, OSPF or RIP can be configured to establish communication between different networks.

Working of Dynamic Routing

In dynamic routing, routers communicate with neighboring routers by exchanging routing updates. These updates contain information about available networks and paths.

When a data packet is sent from one network to another:

1. The router checks its routing table.
2. The routing table contains routes learned dynamically from neighboring routers.
3. The router forwards the packet through the best available path.

If a network change occurs, such as a link failure or addition of a new network, dynamic routing protocols automatically update the routing information without manual intervention.

For example:

- RIP uses hop count to determine the best path.
- OSPF uses cost and shortest path algorithms for route selection.

Importance of Dynamic Routing in Networking

Dynamic routing plays an important role in modern computer networks because it allows routers to communicate and update routing information automatically. It is widely used in enterprise and large-scale networks where manual route configuration is difficult and time-consuming.

Protocols such as RIP and OSPF help maintain reliable communication between networks and improve overall network efficiency.

Learning dynamic routing provides a strong foundation for advanced networking concepts and helps students understand real-world routing mechanisms used in modern network infrastructures.

EXPERIMENT – 08

AIM: Configure DHCP Server in the Network using Packet Tracer software.

Introduction

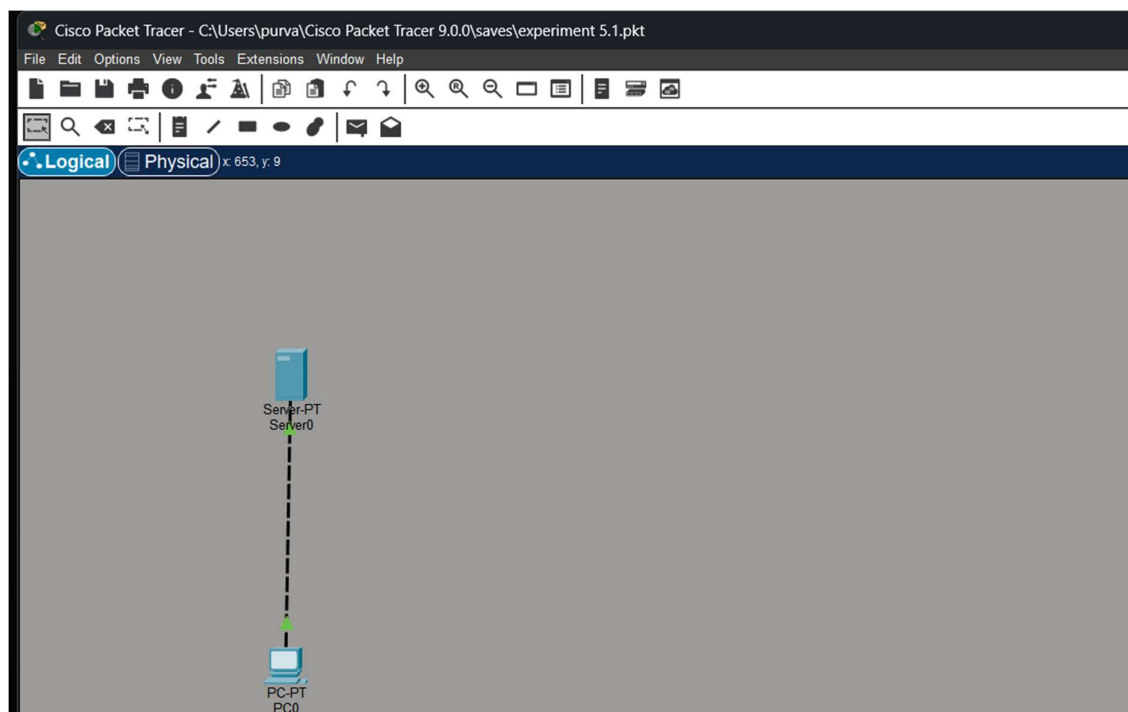
A computer network is a collection of interconnected devices such as computers, switches, routers, and servers that communicate with each other to share data and network resources. In modern networking, automatic IP address assignment plays an important role in reducing manual configuration work. To achieve this, the Dynamic Host Configuration Protocol (DHCP) is widely used.

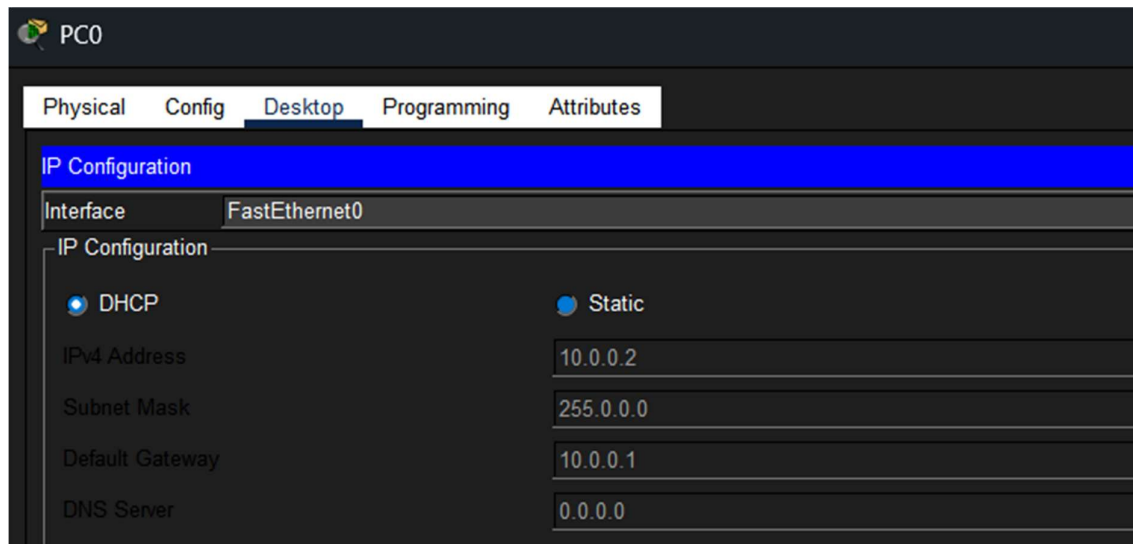
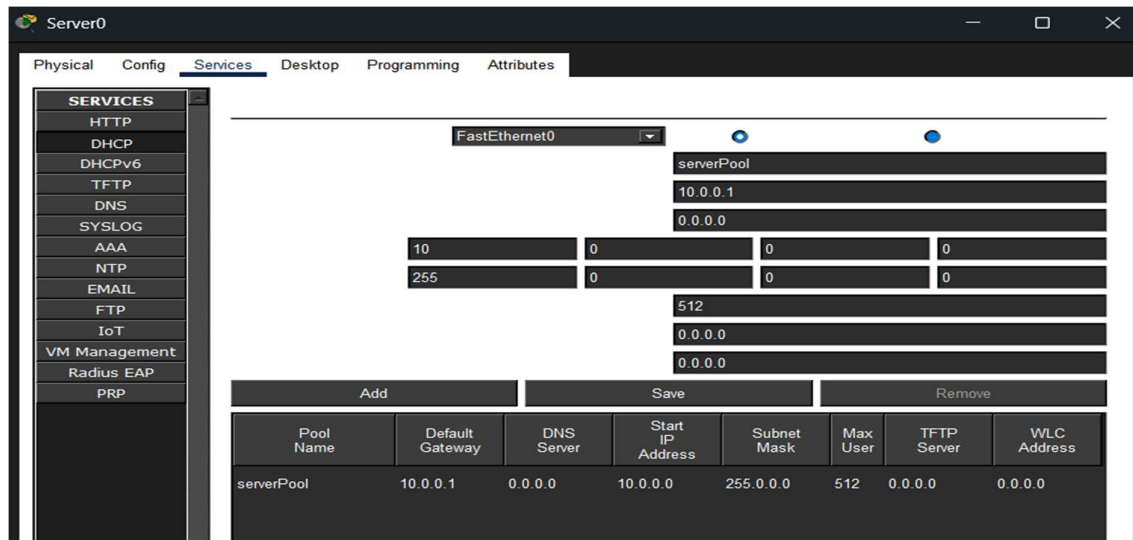
DHCP Server

DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to automatically assign IP addresses and other network configuration details to devices connected to the network. A DHCP server automatically provides:

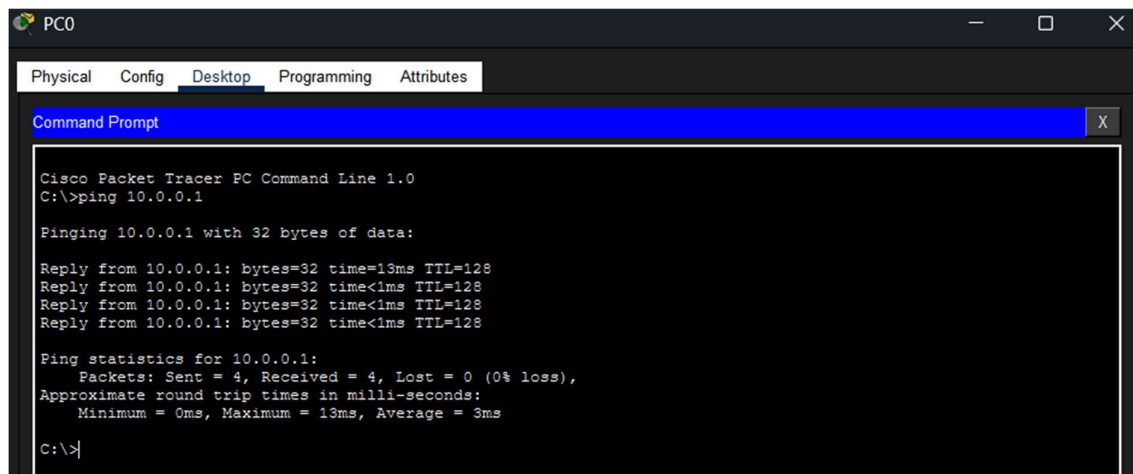
- IP Address
- Subnet Mask
- Default Gateway
- DNS Server Information

This eliminates the need for manually assigning IP addresses to every computer in the network.





OUTPUT:



Network Topology

Network topology refers to the physical or logical arrangement of devices in a computer network. It defines how devices are connected and how communication takes place between them.

In Packet Tracer, users create network topologies by placing networking devices into the workspace and connecting them using appropriate communication cables. In a DHCP network topology:

- PCs act as client devices requesting IP addresses.
- Switches connect devices within the same local network.
- The DHCP Server automatically assigns network configuration to clients.

Working of DHCP

When a computer is connected to the network, it sends a DHCP request to obtain an IP address automatically. The DHCP server receives the request and assigns an available IP address along with other network details.

The DHCP process mainly follows four steps:

1. DHCP Discover – Client searches for a DHCP server.
2. DHCP Offer – Server offers an available IP address.
3. DHCP Request – Client requests the offered IP address.
4. DHCP Acknowledgement – Server confirms the assignment.

This process allows devices to communicate on the network without manual IP configuration.

Importance of DHCP in Networking

DHCP is widely used in modern computer networks because it simplifies network management and reduces administrative workload. It is especially useful in organizations where many devices connect to the network regularly.

Understanding DHCP configuration in Packet Tracer helps students learn automatic network addressing and prepares them for advanced networking concepts and real-world network administration tasks.

EXPERIMENT – 09

AIM: Configure network topology to implement VLANs with trunking using Packet Tracer.

Introduction

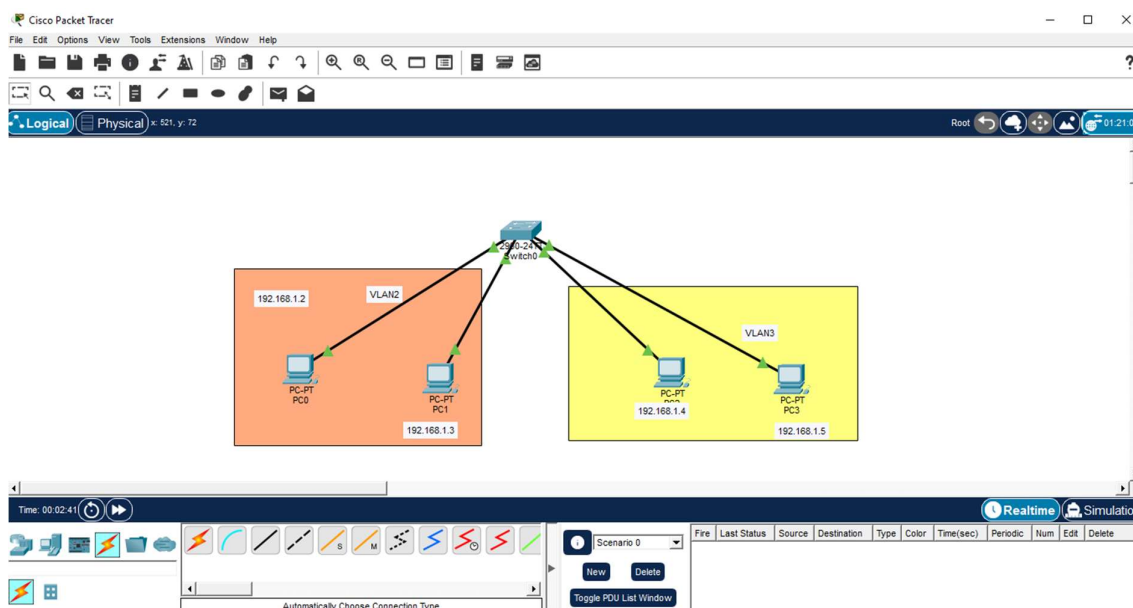
A computer network consists of interconnected devices such as computers, switches, routers, and servers that communicate with each other to exchange information and share resources. In modern networking, network segmentation is important for improving security, reducing traffic congestion, and managing networks efficiently. One of the commonly used techniques for network segmentation is VLAN.

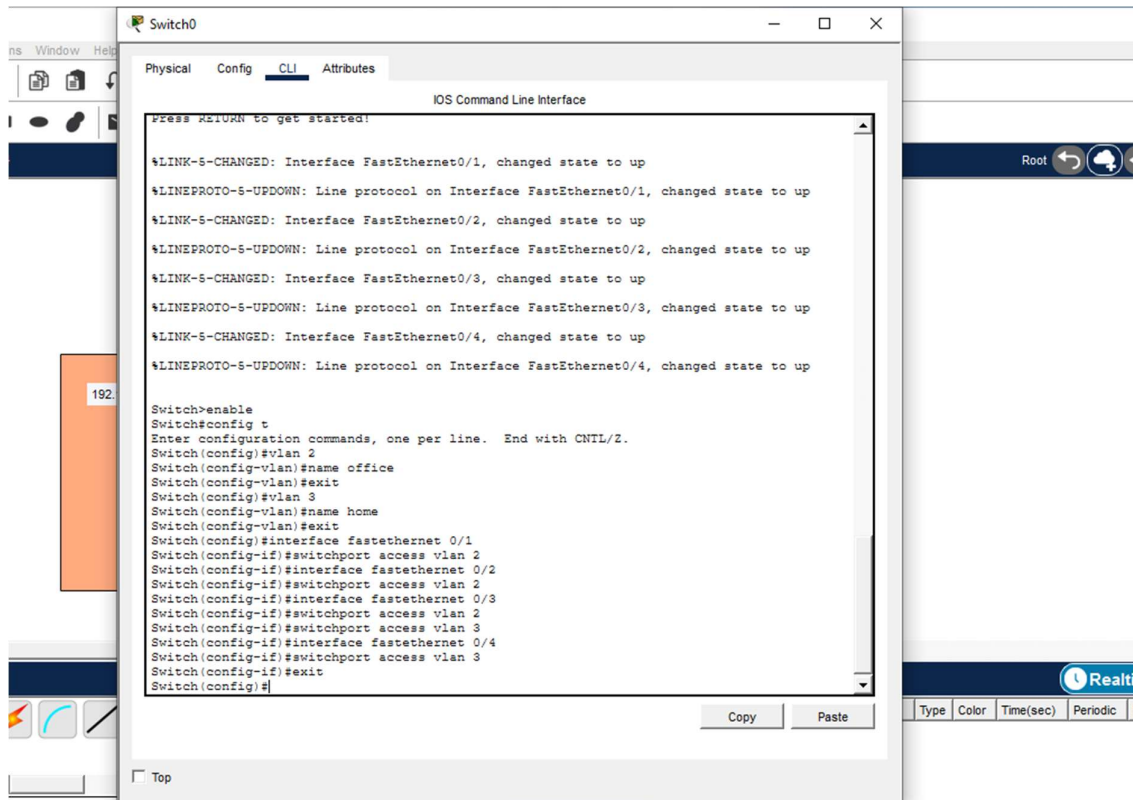
Cisco Packet Tracer is a network simulation software used by students and networking professionals to design, configure, and test network topologies virtually. It helps users understand networking concepts practically by creating simulated network environments.

VLAN (Virtual Local Area Network)

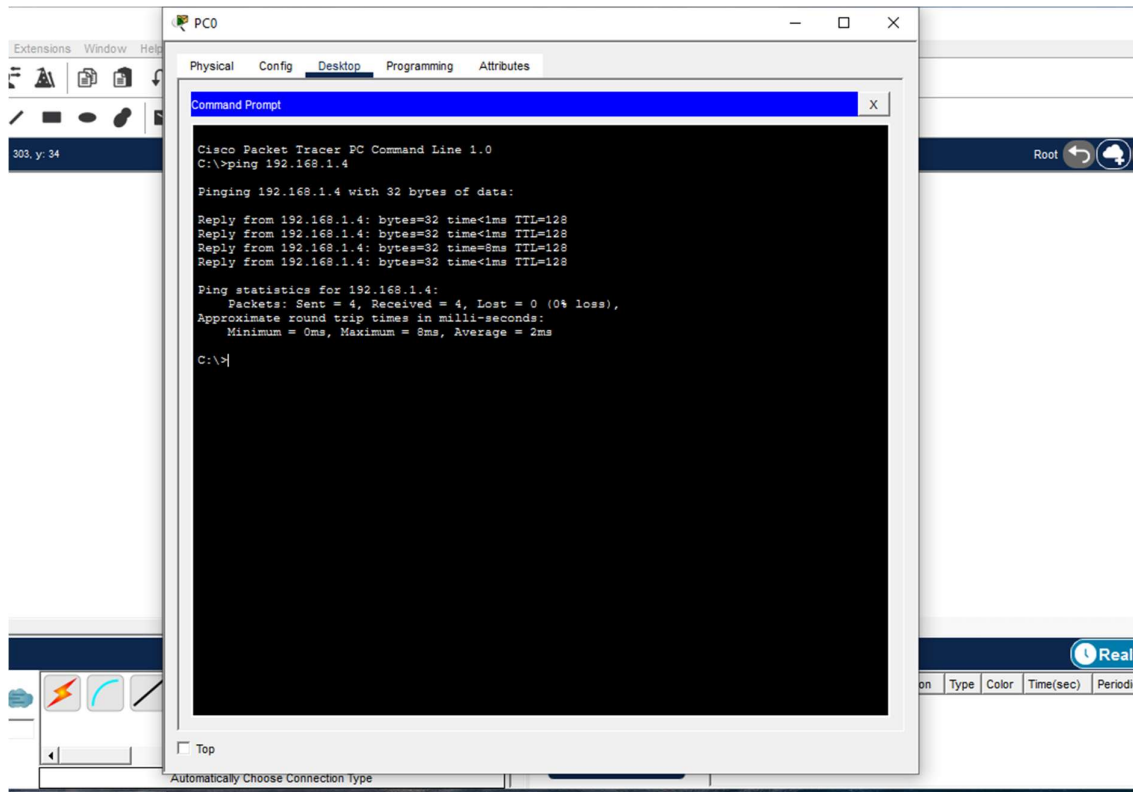
A VLAN (Virtual Local Area Network) is a logical grouping of devices within a network regardless of their physical location. VLANs divide a large network into smaller virtual networks, allowing devices in the same VLAN to communicate with each other as if they were connected to the same physical network.

VLANs improve network performance, security, and management by separating broadcast domains within a switch.





OUTPUT:



Network Topology

Network topology refers to the arrangement or structure of networking devices and their interconnections in a computer network. In Packet Tracer, users can create network topologies by placing devices in the workspace and connecting them using communication cables.

In a VLAN topology with trunking:

- PCs act as end devices connected to different VLANs.
- Switches are used to create and manage VLANs.
- Trunk links are used to carry traffic from multiple VLANs between switches

Importance of VLANs and Trunking in Networking

VLANs and trunking are widely used in organizations to organize departments, improve security, and optimize network performance. They provide flexibility in network design and make large networks easier to manage.

Learning VLAN configuration and trunking in Packet Tracer helps students understand important switching concepts and prepares them for advanced networking technologies used in real-world computer networks.

EXPERIMENT – 10

AIM: Configure network topology and implement dynamic routing protocol such as RIP using Packet Tracer.

Introduction

A computer network is a collection of interconnected devices such as computers, switches, routers, and servers that communicate with each other to exchange data and share resources. In large networks containing multiple routers, efficient communication between different networks requires the use of routing protocols. One of the simplest dynamic routing protocols used in computer networking is RIP.

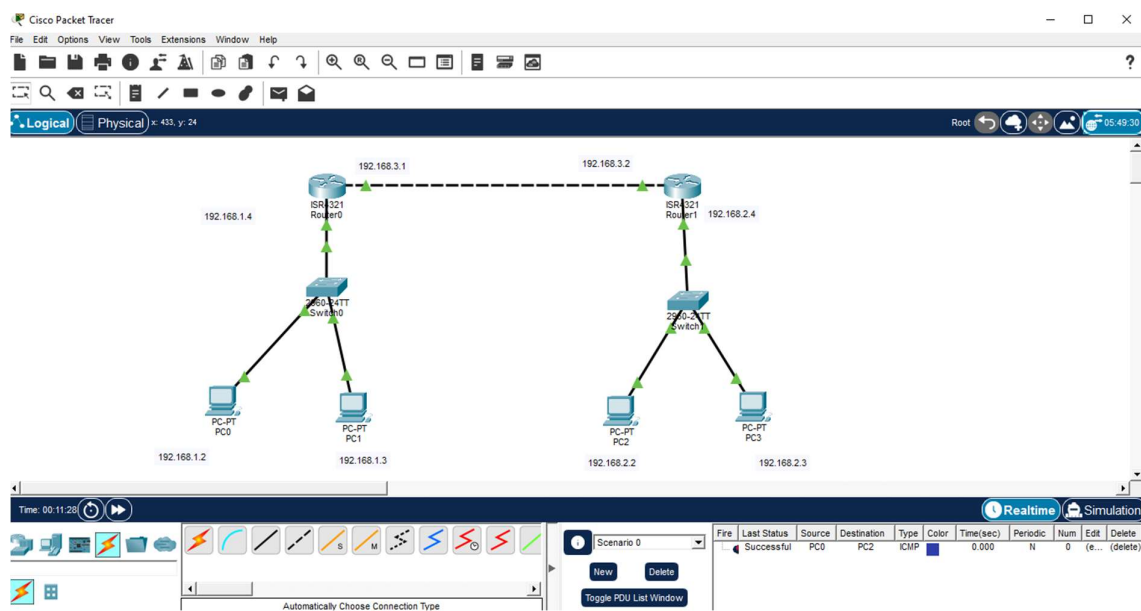
RIP (Routing Information Protocol)

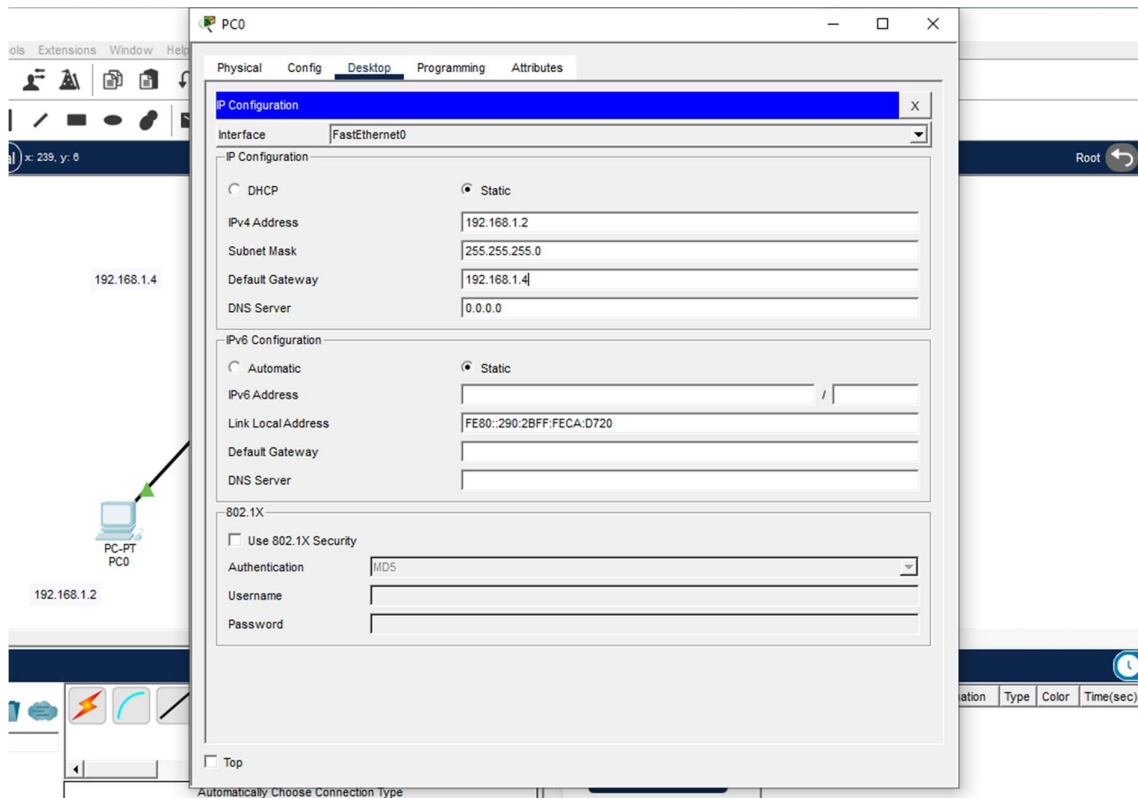
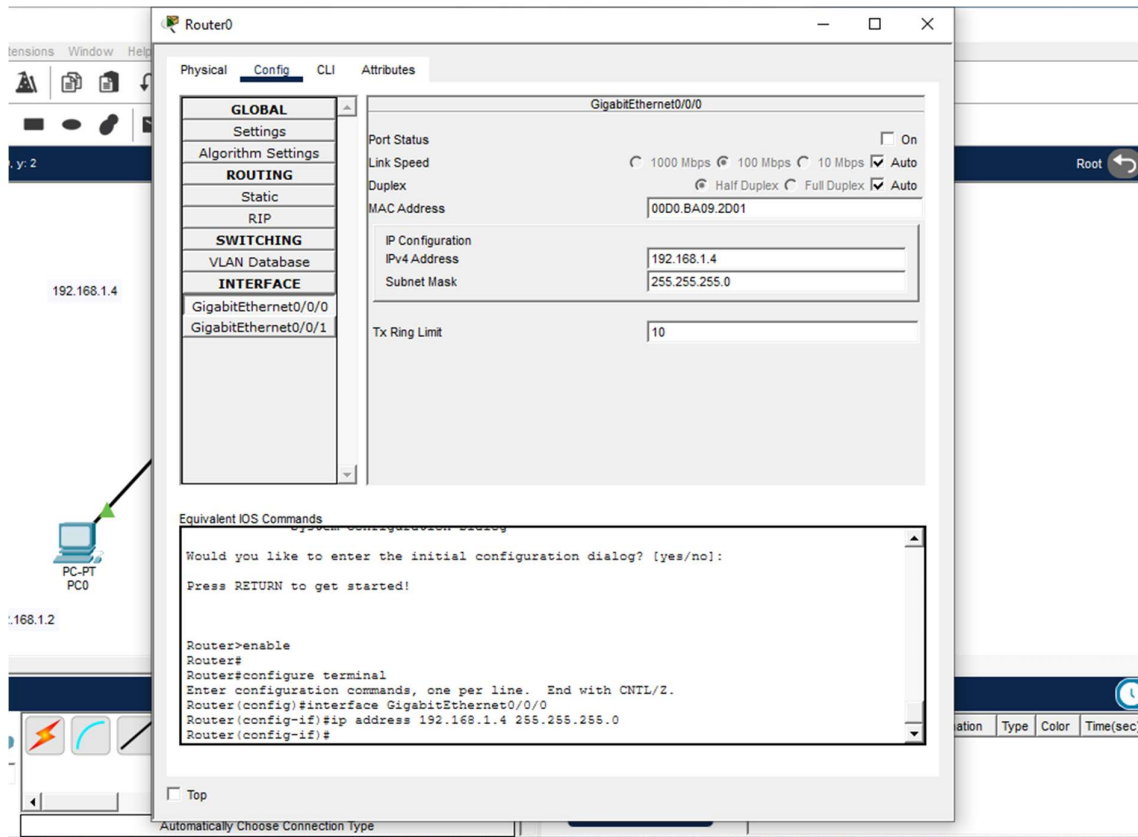
RIP (Routing Information Protocol) is one of the oldest and simplest dynamic routing protocols used in computer networks. RIP uses hop count as the routing metric to determine the best path between networks.

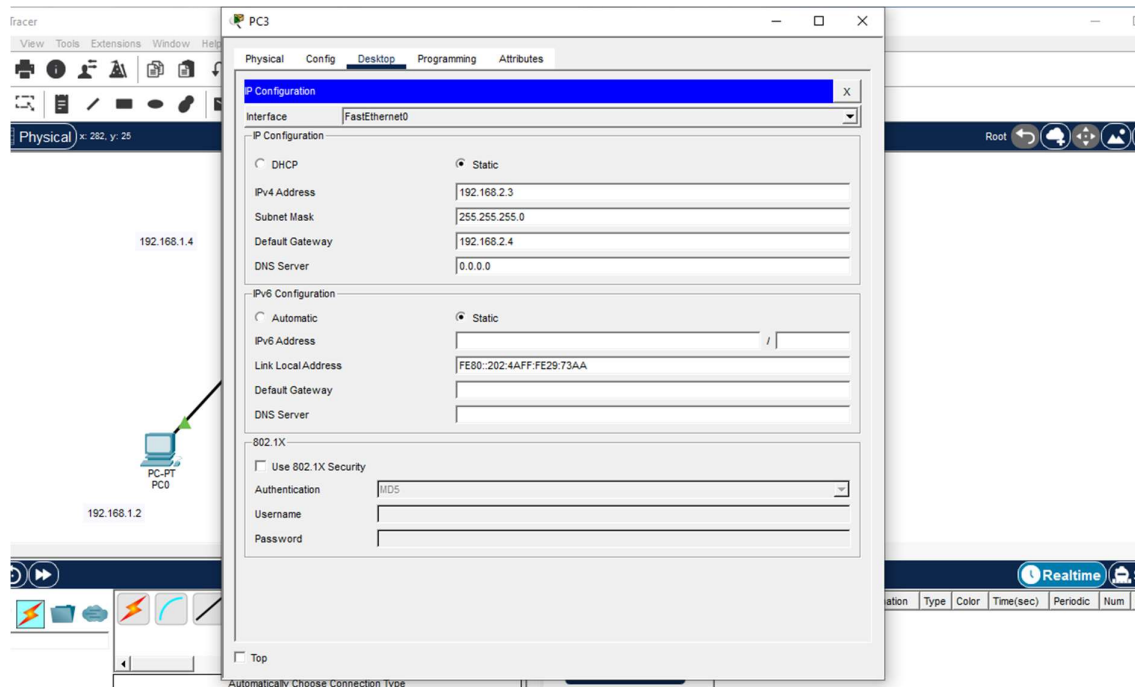
In RIP:

- The maximum hop count allowed is 15.
- A hop count greater than 15 is considered unreachable.
- Routers periodically exchange routing information with neighbouring routers.

RIP operates using the distance-vector routing algorithm.







Network Topology

Network topology refers to the arrangement or structure of networking devices and their connections in a network. In Packet Tracer, users create topologies by placing networking devices into the workspace and connecting them using communication cables.

In a RIP routing topology:

- PCs act as end devices.
- Switches connect devices within the same local network.
- Routers connect multiple networks and exchange routing information using RIP.

Importance of Dynamic Routing and RIP in Networking

Dynamic routing protocols such as RIP are important for managing networks efficiently. They help routers automatically adapt to network changes and maintain communication between multiple networks.

Learning RIP configuration in Packet Tracer helps students understand the basics of dynamic routing and provides a strong foundation for studying advanced routing protocols such as OSPF and EIGRP.